

Track 2 Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

Track 2 is the official offline verification surface for newly introduced TE models. A model or family is not considered verified only because its training campaign metrics improved; it must also be compared on the direction-aware Track 2 curve-reconstruction matrix and reviewed against visual curve evidence.

Current closeout verdict after the wave 2 temporal refresh:

- Track 2 remains the canonical offline model-verification report.
- The wave 2 temporal candidates are verified in the matrix and visual companion reports.
- No temporal candidate is promoted over the current repository-owned tree / hist_gradient_boosting offline baseline.
- The strongest temporal forward candidate is gru_sequence_Fw .
- The strongest temporal backward candidate is lstm_sequence_Bw .
- The strongest temporal global candidate is lstm_sequence_global .
- Future work should treat the temporal models as verified exploratory baselines, not as the accepted offline winner.

Source Package

This official report consolidates these approved `Track 2` artifacts:

- metric matrix:
`doc/reports/analysis/track2/Track 2 Directional Model Comparison.md` ;
- best-model collage report:
`doc/reports/analysis/track2/best_model_collage_report/[2026-05-24]/track2_best_model_collage_report.md` ;
- best-model collage PDF:
`doc/reports/analysis/track2/best_model_collage_report/[2026-05-24]/track2_best_model_collage_report.pdf` ;
- multi-model curve comparison report:
`doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-05-24]/track2_multi_model_curve_comparison_report.md` ;
- multi-model curve comparison PDF:
`doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-05-24]/track2_multi_model_curve_comparison_report.pdf` . Machine-readable and visual validation outputs are retained under:
 - `output/validation_checks/track2_reference_comparison/2026-05-24-22-01-57_track2_full_directional_family_matrix_wave2_temporal_refresh/` ;
 - `output/validation_checks/track2_best_model_collage_report/2026-05-24-22-50-30_track2_best_model_collage_report/` ;
 - `output/validation_checks/track2_multi_model_curve_comparison_report/2026-05-24-23-28-22_track2_multi_model_curve_comparison_report/` .

Verification Rule

The repository treats direction as a first-class verification surface:

Surface	Training or Archive Scope	Evaluation Scope
<code>global</code>	forward and backward together	both directions, reported separately
<code>Fw / forward</code>	forward only	forward curves only
<code>Bw / backward</code>	backward only	backward curves only

The rule applies to paper-reference models, `Track 1` faithful archives, retuned reference archives, `Wave 1` exported models, `Wave 2` temporal registry models, and future `Track 2` campaign candidates.

Pipeline Coverage

Pipeline or Source	Track 2 Role	Current Status	Verification Artifact
recovered original RCIM archive	paper-original forward reference	included	directional matrix
retuned RCIM archive	current paper-derived forward and backward baseline	included	directional matrix and visual reports
<code>Track 1</code> exact paper-faithful bank	source-faithful reproduction evidence	included	directional matrix and visual reports
<code>Wave 1</code> exported static baselines	repository-owned static candidates	included	directional matrix and visual reports
<code>periodic_mlp</code> explicit harmonic campaign	latest family-registry refresh	included in visual refresh	collage and overlay reports

Pipeline or Source	Track 2 Role	Current Status	Verification Artifact
Wave 2 temporal entry campaign	temporal convolution, GRU , and LSTM registry candidates	included	matrix, collage, overlay, PDF
future Track 2 campaigns	new verification candidates	append here when approved	matrix, collage, overlay, PDF

Current Numeric Baselines

Best Composite References

Candidate	Source	Direction	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
paper_original_best_Fw	rcim_original	forward	0.002769	0.002951	6.250	13.827
paper_retuned_best_Fw	rcim_retuned	forward	0.001839	0.002041	4.109	9.866
track1_best_Fw	rcim_track1	forward	0.003014	0.003204	6.819	11.638
paper_retuned_best_Bw	rcim_retuned	backward	0.003675	0.004284	7.572	15.645
track1_best_Bw	rcim_track1	backward	0.005027	0.005212	11.860	48.106

Repository-Owned Static Baselines

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	tree_Fw	0.003053	0.003395	6.731	11.995
backward	tree_Bw	0.003258	0.003651	7.051	14.116
global, forward side	tree_global	0.002998	0.003364	6.590	11.601
global, backward side	tree_global	0.003290	0.003702	7.118	13.703

Wave 2 Temporal Candidates

Direction	Strongest Temporal	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	gru_sequence_Fw		0.003330	0.003762	7.378	13.029
backward	lstm_sequence_Bw		0.003555	0.003985	7.767	14.507
global, combined	lstm_sequence_global		0.003480	0.003903	7.654	12.430

The temporal family is competitive with the non-tree neural/static baselines, but it does not beat the accepted tree baseline in either direction.

Visual Verification Evidence

Best-Model Collage Evidence

The refreshed best-model collage report now includes:

- forward and backward composite reference collages;
- Wave 1 family-best collages;
- explicit-harmonic periodic_mlp refresh collages;
- Wave 2 temporal collages for temporal_convolution , gru_sequence , and lstm_sequence across global , Fw , and Bw . The PDF companion is: doc/reports/analysis/track2/best_model_collage_report/[2026-05-24]/track2_best_model_collage_report.pdf Representative evidence from the refreshed validation output is stored at:
- output/validation_checks/track2_best_model_collage_report/2026-05-24-22-50-30__track2_best_model_collage_report/collages/forward_wave2/gru_sequence_fw.png ;
- output/validation_checks/track2_best_model_collage_report/2026-05-24-22-50-30__track2_best_model_collage_report/collages/backward_wave2/lstm_sequence_bw.png .

Multi-Model Overlay Evidence

The refreshed multi-model curve comparison report now includes dedicated wave 2 overlays and combined reference/tree/temporal overlays for both directions. The combined overlays show the temporal models against the current paper-derived references, Track 1 references, and tree baseline on the same curves.

The PDF companion is
doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-05-24]/track2_multi_model_curve_comparison_report.pdf

Representative evidence from the refreshed validation output is stored at:

- output/validation_checks/track2_multi_model_curve_comparison_report/2026-05-24-23-28-22__track2_multi_model_curve_comparison_report/comparisons/forward_reference_tree_wave2.png ;
- output/validation_checks/track2_multi_model_curve_comparison_report/2026-05-24-23-28-22__track2_multi_model_curve_comparison_report/comparisons/backward_reference_tree_wave2.png .

Campaign Update Ledger

Future Track 2 campaigns must append a row here after the campaign result report, matrix refresh, visual report refresh, and official PDF validation are complete.

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-05-21	periodic_mlp explicit harmonic registry refresh	global , Fw , Bw visual verification	source matrix unchanged; visual registry refresh included	collage and overlay PDFs refreshed	included as visual evidence, not promoted over tree
2026-05-24	Wave 2 temporal-model entry campaign	global , Fw , Bw for TCN, GRU , and LSTM	matrix refreshed from 75 to 84 candidates	collage and overlay PDFs refreshed	verified, but not promoted over tree

Maintenance Contract

For every future `Track 2` model-verification update:

1. refresh `Track 2 Directional Model Comparison.md` when the candidate set or numeric matrix changes;
2. regenerate the best-model collage report when a new family or promoted candidate needs local curve inspection;
3. regenerate the multi-model curve comparison report when relative visual screening changes;
4. append the campaign or update to this report ledger;
5. export and validate this official report PDF;
6. update `doc/running/te_model_live_backlog.md` ;
7. update `Training Results Master Summary.md` when the accepted current best status or campaign interpretation changes.

Closeout Decision

`Track 2` remains closed as the current official offline verification report. The accepted baseline for future work is unchanged:

- forward paper-derived comparison: `paper_retuned_best_Fw` ;
- backward paper-derived comparison: `paper_retuned_best_Bw` ;
- strongest individual paper-reference family evidence:
`rcim_retuned_GBM19_Fw` and `rcim_retuned_GBM19_Bw` ;
- strongest repository-owned static baseline: `tree` / `hist_gradient_boosting` . The `Wave 2` temporal branch is accepted as verified exploratory evidence, not as the promoted offline winner. The practical next branch should either tune a compact temporal/hybrid model against the `tree` baseline or move the strongest verified candidates into deployment-readiness checks before any TwinCAT-facing promotion decision.